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# **Software Requirements Specification**

**for**

## **Squad Sync**

**Version 1.0 approved**

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# 1. Introduction

## 1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to define the functional and non-functional requirements of SquadSync, an AI-powered web application designed to help multiplayer gamers find compatible teammates based on gaming behavior, communication style, preferred roles, and gaming schedules. The platform aims to improve team coordination, reduce mismatched squads, and enhance the overall gaming experience through intelligent matchmaking, gamer profiling, and real-time communication. This document serves as a reference for developers, testers, project supervisors, and stakeholders throughout the software development lifecycle.

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## 1.2 Document Conventions

This document follows the IEEE 830 Software Requirements Specification (SRS) standard. The following conventions are used throughout the document:

- System refers to the SquadSync web application.
  - User refers to a registered gamer using the platform.
  - Admin refers to the administrator responsible for managing the platform.
  - Functional Requirements are represented using the prefix FR.
  - Non-Functional Requirements are represented using the prefix NFR.
  - Machine Learning modules and APIs are referred to using their respective technical names wherever applicable.
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## 1.3 Intended Audience and Reading Suggestions

This document is intended for project supervisors, software developers, machine learning engineers, testers, and future maintainers of the system.

- Project Supervisors can review the project objectives, scope, and overall system architecture.
- Developers should focus on the functional requirements, system features, interfaces, and database design.
- Machine Learning Engineers should refer to the Gamer DNA generation, player classification, and recommendation modules.
- Testers should use the functional and non-functional requirements for validation and testing.
- Future Maintainers can use this document as technical documentation for future enhancements.

It is recommended to read the document sequentially, beginning with the Introduction, followed by the Overall Description, System Features, Interface Requirements, and Non-Functional Requirements.

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## 1.4 Product Scope

SquadSync is an AI-powered web platform developed to simplify team formation in multiplayer games by recommending compatible teammates based on behavioral analysis instead of relying solely on gaming rank. The system introduces the concept of Gamer DNA, which analyzes a player's gaming personality, preferred role, communication style, teamwork, and play schedule through a structured survey. Using Machine Learning algorithms, SquadSync recommends suitable teammates, predicts player roles, and assists in forming balanced gaming squads.

In addition to intelligent matchmaking, the platform provides gamer profile management, friend management, real-time messaging, team dashboards, and gaming news, creating a centralized platform for multiplayer gaming communities. The system demonstrates the integration of Software Engineering, Web Development, Recommendation Systems, and Machine Learning to solve real-world challenges in online gaming.

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## 1.5 References

The development of SquadSync is based on the following standards and technologies:

1. IEEE Std 830-1998 – IEEE Recommended Practice for Software Requirements Specifications.
2. React.js Official Documentation.
3. FastAPI Official Documentation.
4. PostgreSQL Documentation.
5. Scikit-learn Documentation.
6. Socket.IO Documentation.
7. JSON Web Token (JWT) Authentication Standards.
8. REST API Design Guidelines.
9. Research papers on Recommendation Systems, Behavioral Analytics, Explainable AI, and Multiplayer Team Formation.

## 2. Overall Description

### 2.2 Product Functions

SquadSync provides an integrated platform that enables gamers to discover compatible teammates, build balanced squads, communicate effectively, and stay connected with the gaming community. The major functions of the system include:

- User registration, login, and account management.
- Creation and management of personalized gamer profiles.
- Gamer Survey to analyze player behavior, communication style, and gaming preferences and assign roles accordingly.
- Intelligent teammate recommendation based on compatibility, gaming roles, schedules, and player characteristics.
- Smart team formation by suggesting balanced squad compositions and identifying missing roles.
- Friend management, including sending, accepting, and managing friend requests.

- Real-time one-to-one and team communication through an integrated chat system.
- Team dashboard for managing squads, viewing member information, and monitoring compatibility.
- Gaming news portal providing the latest esports news, tournament updates, and game announcements.
- Administrator module for user management, content moderation, and monitoring overall platform activity.

## 2.3 User Classes and Characteristics

The system primarily supports two categories of users based on their access privileges and responsibilities.

### 1. Gamers (Primary Users)

Gamers are the primary users of the platform. They can register, create and manage their gaming profiles, complete the Gamer Survey, receive AI-powered teammate recommendations, send and accept friend requests, create or join teams, communicate through one-to-one and team chat, and view gaming news and updates.

A gamer who creates a team is automatically assigned as the team owner and is provided with additional team management options such as inviting or removing members and managing basic team settings. These are additional permissions rather than a separate user role.

Users are expected to have basic computer skills and familiarity with online multiplayer games.

### 2. Administrator

Administrators are responsible for managing the overall platform. They can manage user accounts, review reported users or content, moderate inappropriate activities, monitor system usage, and maintain the security and integrity of the application.

Administrators have full access privileges that are not available to regular users.

## 2.4 Operating Environment

SquadSync is a web-based application designed to operate across multiple platforms and modern web browsers.

### Hardware Platform

- Client: Desktop, laptop, tablet, or smartphone with internet connectivity.
- Server: Cloud-based server capable of hosting web services, database, and Machine Learning models.

**Operating Systems** - Windows, Mac, Android, iOS.

### Software Environment

- Frontend: React.js, Tailwind CSS
- Backend: FastAPI (Python 3.11+)
- Machine Learning: Python, Scikit-learn, Pandas, NumPy
- Database: PostgreSQL
- Real-Time Communication: Socket.IO / WebSockets
- Authentication: JWT (JSON Web Token)
- Deployment Platforms: Vercel (Frontend), Render (Backend), Supabase (Database)

### **Web Browsers Supported**

- Google Chrome
- Mozilla Firefox
- Microsoft Edge
- Safari

### **Network Requirements**

- Stable internet connection for authentication, AI matchmaking, chat functionality, and gaming news updates.
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## **2.5 Design and Implementation Constraints**

The development of SquadSync AI is subject to the following constraints:

- The application will be developed using React.js for the frontend and FastAPI for backend services.
  - Machine Learning algorithms will be implemented using Python and Scikit-learn.
  - PostgreSQL will be used as the primary relational database.
  - Real-time messaging will be implemented using Socket.IO/WebSockets.
  - User authentication must be secured using JWT.
  - The system should support responsive design for desktop and mobile devices.
  - The application depends on external APIs for gaming news and esports updates.
  - AI recommendations are limited by the quality and completeness of user survey responses.
  - Deployment will use free or educational cloud platforms such as Vercel, Render, and Supabase, which may have usage limitations.
  - Passwords and sensitive user information must be securely encrypted and stored following standard security practices.
  - The system must comply with standard web development practices and RESTful API architecture.
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## **2.6 User Documentation**

The following user documentation will be provided with SquadSync AI:

- Guided steps to account creation, profile setup, Gamer Survey, AI matchmaking, team management, chat, and news features.
  - Administrator Guide describing user management, content moderation, and dashboard functionalities.
  - Installation and Deployment Guide for setting up the application in a development or production environment.
  - API Documentation for backend REST APIs generated using FastAPI (Swagger/OpenAPI).
  - Online Help through tooltips and in-app guidance for important features.
  - FAQ and Troubleshooting Guide covering common issues and solutions.
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## 2.7 Assumptions and Dependencies

### Assumptions

- Users have access to a stable internet connection.
- Users possess basic computer skills and familiarity with online multiplayer games.
- Users provide accurate information while creating profiles and completing the Gamer Survey.
- External gaming news APIs remain available and functional.
- Supported web browsers have JavaScript enabled.
- Cloud hosting services remain accessible during application usage.
- Machine Learning recommendations improve as more user data becomes available.

### Dependencies

- **React.js** and **Tailwind CSS** for the frontend interface.
- **FastAPI** framework for backend API development.
- **Python**, **Scikit-learn**, **Pandas**, and **NumPy** for AI-based matchmaking.
- **PostgreSQL** for persistent data storage.
- **Socket.IO/WebSockets** for real-time chat functionality.
- **JWT** for secure authentication and authorization.
- External gaming news APIs for esports news, tournament updates, and patch notes.
- Cloud deployment platforms including **Vercel**, **Render**, and **Supabase**.
- Internet connectivity for authentication, AI recommendations, chat services, and live news updates.

## 3. External Interface Requirements

### 3.1 User Interfaces

The SquadSync AI platform features a responsive web interface built with **React.js** and **Tailwind CSS**. This design ensures that gamers can navigate features intuitively while maintaining high accessibility and consistency across various devices.

The system provides the following specialized user interfaces:

### **Authentication Interface**

This module secures entry to the platform through several essential functions:

- New user registration and secure login protocols.
- Credential recovery through a forgot password mechanism.
- Session security via **JWT**-based authentication.
- Real-time input validation and user-friendly error messaging.

### **User Dashboard**

Acting as the primary hub after authentication, the dashboard provides a personalized overview and streamlined navigation to core modules:

- Summarized views of the Gamer Profile and characteristics.
- High-level lists of AI-generated teammate suggestions.
- Alerts for system notifications and recent team activities.

### **Gamer Profile Interface**

Users utilize this interface to define and manage their professional gaming identity:

- Configuration of game preferences, competitive ranks, and primary roles.
- Selection of regional settings and supported languages.
- Management of active gaming schedules and preferred communication channels.

### **AI Gamer Survey**

This specialized questionnaire gathers psychological and behavioral data essential for matchmaking:

- Assessment of communication styles and teamwork aptitude.
- Evaluation of leadership qualities, strategic thinking, and competitive drive.
- Analysis of general gaming habits to refine the unique profile.

The platform processes these inputs to establish the user's distinct Gamer Profile.

### **AI Matchmaking Interface**

This view presents the outputs from the recommendation engine, highlighting potential teammates based on several criteria:

- Numerical compatibility scores and explainable justifications for each suggestion.
- Visibility into shared schedules and complementary gaming roles.
- Advanced filtering options to refine teammate discovery.

### **Friend Management Interface**

Users build their social network within the application through this module:

- Global player search and outgoing request management.
- Review and response to incoming friend invitations.
- Active friend list viewing and contact removal capabilities.

### **Team Dashboard**

The dashboard offers a comprehensive overview of squad metrics and composition:

- Roster management showing members and their respective assigned roles.
- Collective squad compatibility and calculated average ranks.
- Identification of overlapping schedules for coordinated gameplay.

### **Real-Time Chat Interface**

The messaging system facilitates instantaneous communication across several channels:

- Direct one-to-one messaging and integrated group team chats.
- Live visibility of online status and instant message alerts.
- Persistent access to historical chat logs.

### **Gaming News Portal**

This portal serves as a curated feed for the latest community information:

- Aggregated esports journalism and official tournament schedule updates.
- Direct feeds for game patch notes and industry announcements.

### **Admin Dashboard**

Privileged users manage the health and integrity of the system through this interface:

- Comprehensive user account oversight and report moderation.
- Real-time platform monitoring and analytics on system statistics.
- Curation and oversight of the gaming news module.

The application maintains uniform typography, color palettes, and navigation components across all modules. Standardized interface elements—including authentication, data management, and social interaction buttons—are deployed consistently. Furthermore, the system displays validation feedback and success alerts in a transparent, accessible format.

## **3.2 Hardware Interfaces**

SquadSync AI functions as a web-based service, eliminating the need for custom hardware. It is compatible with standard internet-connected devices, including:

- Laptops and desktops with standard mouse or keyboard inputs.
- Mobile phones and tablets featuring touchscreen navigation.
- Displays supporting a minimum 1366 × 768 resolution.
- Network connectivity provided by Wi-Fi or Ethernet.

Communication with hardware is facilitated via standard browser APIs and OS drivers. While the server manages AI computations and database logic, the client device handles only the user interface rendering and local interactions.

### 3.3 Software Interfaces

The system integrates several software frameworks to deliver the core matchmaking, messaging, and data security features.

#### Operating System

Supported environments include **macOS**, **Linux**, and **Windows**, accessed through modern web browsers.

#### Frontend

Developed using **React.js**, **Tailwind CSS**, HTML5, and CSS3, leveraging modern JavaScript (ES6+).

#### Backend

API services are implemented using the **FastAPI** framework on **Python**.

#### Machine Learning Module

The AI engine utilizes **Scikit-learn**, **Pandas**, and **NumPy** to execute clustering and similarity algorithms for teammate recommendations.

#### Database

Persistent storage for accounts, survey results, squad information, and activity logs is managed via **PostgreSQL**.

#### Authentication

Secure sessions and resource authorization are maintained through the use of **JWT**.

#### Real-Time Communication

Instant messaging is powered by **Socket.IO** and WebSockets, enabling low-latency communication.

#### External APIs

Integrations with third-party news services provide esports coverage and game update data. Component interactions are structured around RESTful APIs using JSON, ensuring a secure and modular architecture.

### 3.4 Communications Interfaces

Reliable connectivity is required between users and the underlying services to maintain real-time functionality. Key specifications include:

- Use of HTTP/HTTPS for all standard client-server traffic.
- RESTful architecture and JSON data formats for matchmaking and core services.
- **Socket.IO/WebSockets** for synchronization of chat and team updates.
- Network security enforced by **SSL/TLS** encryption and **JWT** authorization.
- Integration of external news APIs for live esports content.
- Standardized **HTTP status codes** for robust error handling.

## 4. System Features

### 4.1 User Authentication & Authorization

#### Description

The User Authentication and Authorization module provides secure access to the SquadSync platform by managing user registration, login, and authentication. It ensures that only authorized users can access the application while protecting user credentials and personal information. This module forms the foundation of the system by maintaining secure user sessions and preventing unauthorized access.

#### Functionalities

- User Registration
  - Secure Login
  - Password Encryption
  - JWT-based Authentication
  - Session Management
  - Logout Functionality
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### 4.2 Gamer Profile Management

#### Description

The Gamer Profile Management module allows users to create and maintain their personalized gaming profiles. It stores important gaming information such as preferred games, gaming rank, preferred roles, language, region, and gaming schedule. These profiles serve as the primary source of information for the AI recommendation engine to provide accurate teammate suggestions.

#### Functionalities

- Create and Update Gamer Profile
  - Upload Profile Picture
  - Select Preferred Games
  - Add Gaming Rank
  - Choose Preferred Roles
  - Set Gaming Availability
  - Manage Profile Visibility
-

### 4.3 Gamer DNA Analysis Engine

#### Description

The Gamer DNA Analysis Engine is the core feature of SquadSync. It collects player behavior, gaming preferences, and decision-making patterns through an interactive survey. Based on the responses, the system generates a unique Gamer DNA profile that represents the player's gaming personality, communication style, teamwork, leadership, strategic thinking, and playstyle. This profile acts as the foundation for all AI-powered recommendations.

#### Functionalities

- Interactive Gaming Behavior Survey
  - Gamer DNA Generation
  - Gaming Personality Analysis
  - Gamer DNA Card Generation
  - Gaming Trait Visualization
  - Survey Progress Tracking
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### 4.4 AI Gamer Role Classification

#### Description

This module utilizes Machine Learning to analyze the Gamer DNA profile and identify the most suitable gaming role for each player. Instead of relying solely on self-selected roles, the system predicts the player's strengths and recommends both primary and secondary gaming roles based on behavioral analysis.

#### Functionalities

- Predict Primary Gamer Role
  - Predict Secondary Gamer Role
  - Gamer Archetype Classification
  - Role Confidence Score
  - Playstyle Recommendation
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### 4.5 AI Matchmaking & Recommendation Engine

#### Description

The AI Matchmaking and Recommendation Engine intelligently recommends compatible teammates by analyzing Gamer DNA, gaming roles, communication preferences, gaming schedules, language, region, and experience level. The system generates compatibility scores and provides explainable recommendations that help users understand why a particular teammate has been suggested.

#### Functionalities

- Compatibility Score Generation
  - Intelligent Teammate Recommendation
  - Explainable AI Recommendations
  - Schedule-Based Matching
  - Region and Language Matching
  - Similar Playstyle Detection
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### 4.6 Smart Team Formation

#### Description

The Smart Team Formation module creates balanced gaming squads by evaluating player compatibility and role distribution. It ensures that teams consist of complementary gaming roles and minimizes role conflicts by suggesting suitable replacements whenever required.

#### Functionalities

- Team Creation
  - Balanced Squad Formation
  - Role Distribution Analysis
  - Missing Role Detection
  - Team Compatibility Analysis
  - Squad Recommendations
- 

### 4.7 Friend Management

#### Description

The Friend Management module enables users to build and manage their gaming network. Players can send and receive friend requests, maintain long-term gaming connections, and quickly invite friends to join teams and gaming sessions.

#### Functionalities

- Send Friend Requests
  - Accept or Reject Friend Requests
  - Friends List Management
  - Remove Friends
  - View Online Status
- 

## 4.8 Real-Time Communication

### Description

The Real-Time Communication module provides instant messaging services that enable gamers to coordinate before and during gameplay. Users can communicate individually or within teams using an integrated chat system that supports seamless interaction.

### Functionalities

- One-to-One Chat
  - Team Chat
  - Real-Time Messaging
  - Chat History
  - Online User Status
- 

## 4.9 Gaming News Portal

### Description

The Gaming News Portal provides users with the latest updates from the gaming industry, including esports tournaments, patch notes, game releases, and major announcements. The content is dynamically fetched through external news APIs to ensure users receive up-to-date information.

### Functionalities

- Latest Gaming News
  - Esports Updates
  - Patch Notes
  - Tournament Announcements
  - Personalized News Feed
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#### 4.10 Team Dashboard

##### Description

The Team Dashboard provides a centralized overview of the gaming squad by displaying team members, assigned roles, compatibility scores, gaming schedules, and overall team statistics. It helps players monitor and coordinate their teams efficiently.

##### Functionalities

- Team Overview
  - Team Member Information
  - Role Distribution
  - Team Compatibility Score
  - Shared Gaming Schedule
  - Team Statistics
- 

#### 4.11 Admin Dashboard

##### Description

The Admin Dashboard allows administrators to monitor and manage the SquadSync platform efficiently. It provides tools for managing users, moderating content, handling reports, monitoring system activity, and maintaining platform integrity.

##### Functionalities

- User Management
- Platform Monitoring
- User Report Management
- Content Moderation
- News Management
- Platform Analytics

### 5. Other Nonfunctional Requirements

#### 5.1 Performance Requirements

**NF-1.1:** Once the application is fully responsive, standard interface pages such as the dashboard and news feed must load within a 2–3 second window.

**NF-1.2:** Due to the use of free-tier cloud services, a 60-second resumption delay may occur after periods of inactivity; this is a known environment constraint rather than a technical defect.

**NF-1.3:** The AI Matchmaking Engine is required to deliver teammate suggestions within 5–8 seconds, accounting for the limited processing power of the hosting platform.

**NF-1.4:** Delivery latency for real-time messages across one-to-one and group chats should remain under 1 second during standard operation.

**NF-1.5:** Support for 10–30 simultaneous users is expected, reflecting the application's status as an academic project rather than a commercial deployment.

**NF-1.6:** Database structures must be refined to stay within free-tier capacity by eliminating redundant logs and verbose data once processing is finalized.

**NF-1.7:** Expanding infrastructure for larger user bases or higher data throughput is considered outside the current project scope and reserved for future development.

## **5.2 Safety Requirements**

**NF-2.1:** Since automated backups are not guaranteed by the host, manual or scripted data preservation for profiles and chat history must be conducted at key milestones.

**NF-2.2:** Proactive measures will be taken to ensure the backend and database remain active during scheduled evaluations to prevent suspension due to idle states.

**NF-2.3:** If the AI engine or third-party APIs fail, the system must degrade gracefully, maintaining the availability of core features like profile management and chat.

**NF-2.4:** To manage security without paid resources, a cooling-off period will be triggered after multiple unsuccessful login attempts to deter unauthorized access.

## **5.3 Security Requirements**

**NF-3.1:** Accessing secure system resources requires a valid JWT, which is generated only upon a verified user login.

**NF-3.2:** Plaintext password storage is prohibited; all credentials must be secured using industry-standard hashing techniques like bcrypt before being saved.

**NF-3.3:** All traffic between the client and server must be encrypted via HTTPS to protect data while it is in transit.

**NF-3.4:** Role-based permissions will be enforced to distinguish between standard users and administrators, limiting sensitive functions to authorized personnel.

**NF-3.5:** API endpoints must validate and sanitize all user input to mitigate risks associated with SQL injection and cross-site scripting (XSS).

**NF-3.6:** Configuration secrets and API keys must be kept in environment variables and excluded from the version control repository.

## **5.4 Software Quality Attributes**

**Reliability:** The application is designed to be stable during evaluation phases, with performance expectations aligned with its academic purpose rather than production-grade SLAs.

**Usability:** The interface will utilize a coherent design language, allowing new users to navigate the survey and teammate recommendations intuitively.

**Maintainability:** Code will be structured into modular blocks for matchmaking and chat, facilitating updates without disrupting independent sections of the platform.

**Portability:** The frontend will support responsive viewing across browsers, while the backend will maintain consistency across various hosting environments.

**Explainability:** Teammate recommendations will include human-readable justifications, such as shared gaming roles or schedules, providing context beyond simple scores.

**Cost Constraint:** The entire application must be maintainable at no financial cost, adhering to its scope as a non-commercial student deliverable.

## **5.5 Business Rules**

**BR-1:** Users are required to finalize the Gamer Survey to become eligible for any AI-powered matchmaking services.

**BR-2:** Networking and friend requests are restricted between accounts where a block or report has been previously initiated.

**BR-3:** Access to team-specific dashboards is strictly limited to the current members of that particular squad.

**BR-4:** Administrative access will be granted manually by the developers rather than via public registration, fitting the academic context of the project.

**BR-5:** Teammate suggestions will be updated dynamically following changes to a user's profile or survey data rather than on a fixed cycle.

**BR-6:** Flagged content or reported profiles will undergo administrator review within a 48-hour window before any moderation actions are applied.